

CHAPTER 11

THERMAL PROPERTIES

& TEMPERATURE

2.2 Thermal properties and temperature

2.2.1 Thermal expansion of solids, liquids and gases

- 1 Explain applications and consequences of thermal expansion in the context of common examples, including the liquid-in-glass thermometer
- 2 Explain, in terms of the motion and arrangement of particles, the thermal expansion of solids, liquids and gases, and state the relative order of magnitudes of the expansion of solids, liquids and gases
- 3 Convert temperatures between kelvin and degrees Celsius; recall and use the equation
 $T \text{ (in K)} = \theta \text{ (in } ^\circ\text{C)} + 273$

2.2.2 Specific heat capacity

- 1 Know that an increase in the temperature of an object increases its internal energy
- 2 Describe an increase in temperature of an object in terms of an increase in the average kinetic energies of all of the particles in the object
- 3 Define specific heat capacity as the energy required per unit mass per unit temperature increase; recall and use the equation

$$\text{specific heat capacity} = \frac{\text{change in energy}}{\text{mass} \times \text{change in temperature}}$$

$$c = \frac{\Delta E}{m\Delta\theta}$$

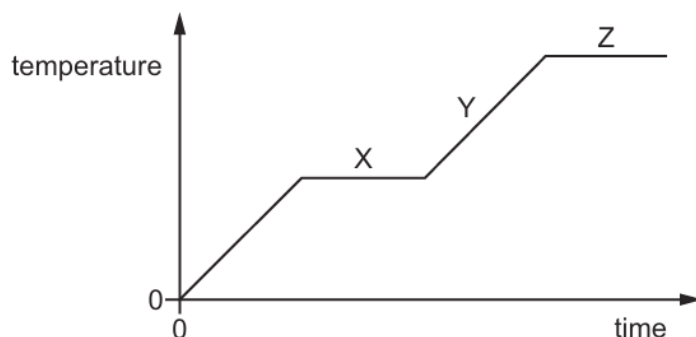
- 4 Describe experiments to measure the specific heat capacity of a solid and of a liquid

2.2.3 Melting, boiling and evaporation

- 1 Describe melting, solidification, boiling and condensation in terms of energy transfer without a change in temperature
- 2 Know the melting and boiling temperatures for water at standard atmospheric pressure
- 3 Describe the differences between boiling and evaporation
- 4 Describe evaporation in terms of the escape of more energetic particles from the surface of a liquid
- 5 Describe how temperature, surface area and air movement over a surface affect evaporation
- 6 Explain how evaporation causes cooling
- 7 Describe latent heat as the energy required to change the state of a substance and explain it in terms of particle behaviour and the forces between particles

1. O/N/23/11/Q15

A graph of temperature against time is shown for a material being heated from its solid state.

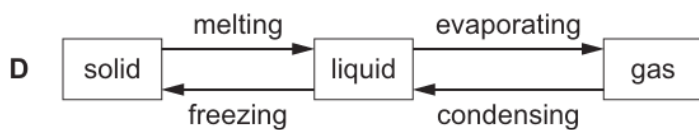
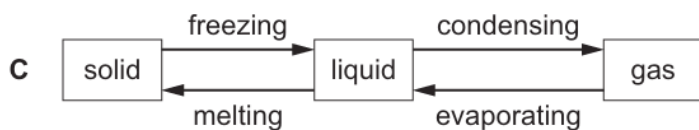
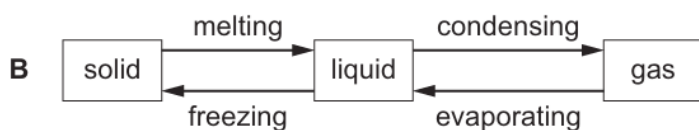
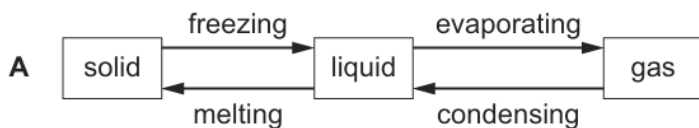


Which row describes what is happening at X, Y and Z?

	X	Y	Z
A	boiling	solid becomes hotter	melting
B	liquid becomes hotter	boiling	gas becomes hotter
C	solid becomes hotter	melting	liquid becomes hotter
D	melting	liquid becomes hotter	boiling

2. O/N/23/12/Q17

Which diagram shows the names of the changes of state that correspond to the directions of the arrows?



3. O/N/23/12/Q18

A sealed cylinder containing gas is left in direct sunlight.

Eventually the cylinder explodes.

How is this explained in terms of the behaviour of the gas particles?

- A Each particle gets bigger and they collide with each other more often.
- B Each particle gets bigger and they collide harder with the cylinder.
- C The particles gain kinetic energy and hit the cylinder harder and more often.
- D The particles lose kinetic energy and they hit the cylinder harder and more often.

4. O/N/23/12/Q20

Pot X and pot Y are both in a cool room.

Pot X contains hot water below its boiling point.

Pot Y is identical to pot X but contains water that is kept boiling by a hotplate underneath it.



What happens to the average kinetic energy of the particles in the liquid water in pot X and to the average kinetic energy of the particles in the liquid water in pot Y?

	kinetic energy of liquid particles in X	kinetic energy of liquid particles in Y
A	decreases	increases
B	decreases	does not change
C	stays the same	increases
D	stays the same	does not change

5. M/J/23/11/Q16

An incomplete statement about latent heat is given.

Latent heat is the1..... required to overcome the bonds between molecules and change the2..... of a substance.

Which words are inserted into gap 1 and gap 2 to correctly complete the sentence?

	gap 1	gap 2
A	energy	state
B	force	state
C	energy	temperature
D	force	temperature

6. M/J/23/11/Q17

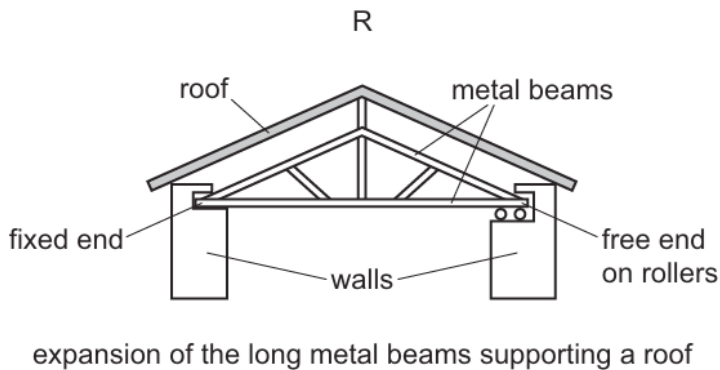
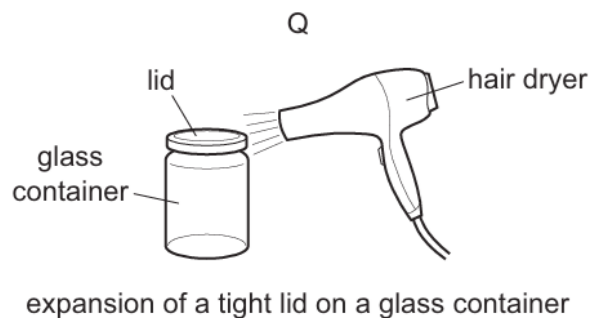
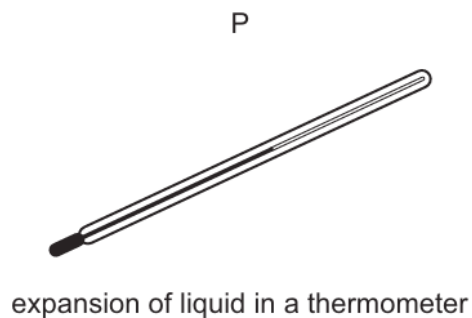
A cold solid is placed on top of a hot solid. Thermal energy is transferred from the hot solid to the cold one.

What is the explanation for this?

- A** A hot solid expands, so its particles will move further apart.
- B** Energy is transferred from one particle to the next.
- C** Heat always rises.
- D** Molecules are free to move randomly through the solids.

7. M/J/23/12/Q16

The diagrams show examples of thermal expansion.



In which diagrams is the thermal expansion useful?

- A** P, Q and R **B** P and Q only **C** P only **D** Q and R only

8. O/N/22/11/Q20

A piece of iron of mass m is placed in a mixture of ice and water.

Its temperature decreases from θ_1 to θ_2 .

How much thermal energy is lost by the piece of iron?

- A** $m \times \text{specific heat capacity of iron} \times (\theta_1 - \theta_2)$
- B** $m \times \text{specific heat capacity of iron} \times \theta_2$
- C** $m \times \text{heat capacity of piece of iron} \times (\theta_1 - \theta_2)$
- D** $m \times \text{heat capacity of piece of iron} \times \theta_2$

9. O/N/22/11/Q21

A kilogram of a substance has a fixed shape.

Thermal energy supplied to the substance weakens the forces of attraction between its molecules.

What is happening?

- A** A gas is being heated.
- B** A liquid is at its boiling point.
- C** A liquid is evaporating.
- D** A solid is at its melting point.

10. O/N/22/11/Q22

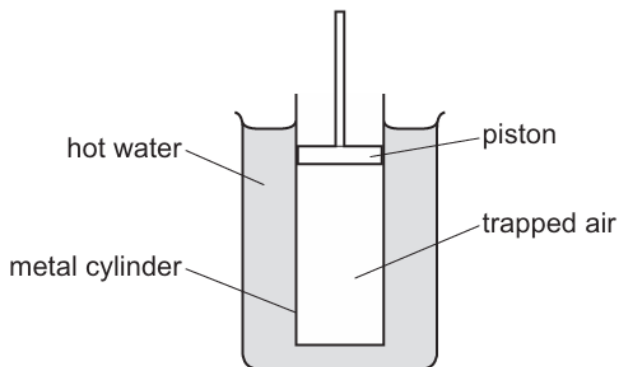
Which conditions produce the greatest rate of evaporation from a liquid?

- A** high temperature, draught and large surface area of liquid
- B** high temperature, no draught and large surface area of liquid
- C** low temperature, draught and small surface area of liquid
- D** low temperature, no draught and small surface area of liquid

11. O/N/22/12/Q21

Air is trapped in a metal cylinder by a piston. The piston is free to move and the trapped air is at atmospheric pressure.

The cylinder is in hot water.



The cylinder is taken out of the water and left to cool.

What happens to the mass of the air in the cylinder and its pressure as it cools?

	mass of air	pressure of air
A	does not change	does not change
B	does not change	changes
C	changes	does not change
D	changes	changes

12. M/J/22/11/Q17

What is the usual symbol and what is the unit for the heat capacity of an object?

	symbol	unit
A	C	$\text{J}/^{\circ}\text{C}$
B	C	$\text{J}/(\text{kg } ^{\circ}\text{C})$
C	H	$\text{J}/^{\circ}\text{C}$
D	H	$\text{J}/(\text{kg } ^{\circ}\text{C})$

13. M/J/22/12/Q20

Steam at 110°C condenses on a surface to form water droplets at 100°C .

What happens after the steam comes into contact with the surface?

- A** The molecules slow down and absorb energy from the surroundings.
- B** The molecules slow down and emit energy to the surroundings.
- C** The molecules stay at the same speed and absorb energy from the surroundings.
- D** The molecules stay at the same speed and emit energy to the surroundings.

14. M/J/22/12/Q21

Which list shows the states of matter in order of expansion from smallest to greatest for the same temperature rise?

- A** gas → liquid → solid
- B** liquid → solid → gas
- C** solid → gas → liquid
- D** solid → liquid → gas

15. M/J/22/12/Q22

Which row describes the boiling and the evaporation of a liquid?

	boiling	evaporation
A	bubbles form throughout the liquid	only occurs at one temperature
B	bubbles form throughout the liquid	produces cooling
C	occurs at any temperature	only occurs at one temperature
D	occurs at any temperature	produces cooling

16. M/J/22/12/Q23

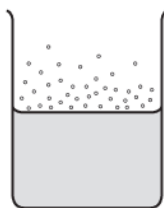
A hairdryer is used to blow air across the surface of water in a beaker so that the water evaporates.

What increases the rate of evaporation of the water?

- A** decreasing the speed of the air from the hairdryer
- B** decreasing the mass of the water in the beaker
- C** increasing the surface area of the water by using a wider beaker
- D** increasing the volume of the water in the beaker

17. O/N/21/11/Q16

The diagram shows a room temperature liquid in a beaker.



What reduces the rate of loss of liquid by evaporation?

- A** blowing air across the top of the beaker
- B** heating the liquid
- C** putting a lid on the beaker
- D** transferring the liquid to a beaker with a larger diameter

18. O/N/21/12/Q18

Solar panels are used to heat water with a mass of 500 kg.

The total area of the solar panels is 10 m^2 and the average power output from each square metre of the panels is 300 W. The specific heat capacity of water is $4200 \text{ J/(kg } ^\circ\text{C)}$.

What is the increase in the water temperature after 8.0 hours? (Assume there is no heat loss.)

- A** 0.69°C **B** 4.1°C **C** 5.1°C **D** 41°C

19. O/N/21/12/Q20

The temperatures of the water in four beakers are different and areas of the surfaces of the water are also different.

In which beaker is the rate of evaporation of the water greatest?

	temperature of water / $^\circ\text{C}$	surface area of water / cm^2
A	20	50
B	20	100
C	25	50
D	25	100

20. M/J/21/11/Q18

What is the *heat capacity* of a body?

- A** the amount of thermal energy that the body can absorb without melting
- B** the amount of thermal energy required to raise the temperature of the body by 1.0°C
- C** the amount of thermal energy required to raise the temperature of 1.0 kg of the body by 1.0°C
- D** the amount of thermal energy required to raise the temperature of 1.0 m^3 of the body by 1.0°C

21. M/J/21/11/Q19

Which statement about water is correct?

- A** At the boiling point, water vapour molecules have the same kinetic energy as liquid water molecules.
- B** Evaporation occurs only at the boiling point.
- C** Water molecules become heavier when water freezes.
- D** Water molecules lose all of their kinetic energy when water freezes.

22. M/J/21/12/Q21

Which statement about the thermal expansion of solids, liquids and gases is correct?

- A** Liquids do not expand.
- B** Liquids expand more than gases but less than solids.
- C** Liquids expand more than solids but less than gases.
- D** Liquids expand the same amount as solids but less than gases.

23. O/N/20/11/Q21

What is a unit for heat capacity?

- A** J
- B** J/g
- C** J/°C
- D** J/(g °C)

24. O/N/20/11/Q22

A block of aluminium is heated.

What happens to the kinetic energy and to the potential energy of the molecules?

	kinetic energy	potential energy
A	increase	increase
B	increase	decrease
C	decrease	increase
D	decrease	decrease

25. O/N/20/11/Q23

Which statement about the melting of ice and the boiling of water is correct?

- A** Both processes are accompanied by a decrease in volume.
- B** Both processes involve energy transfer and a temperature change.
- C** Both processes involve the absorption of latent heat.
- D** Both processes result in the increase of intermolecular forces.

26. O/N/20/11/Q24

A substance has a melting point of -17°C and a boiling point of 117°C .

In which state does the substance exist at -10°C and at 110°C ?

	at -10°C	at 110°C
A	solid	liquid
B	solid	gas
C	liquid	liquid
D	liquid	gas

27. M/J/20/12/Q24

Thermal energy is used to turn water at 100 °C to steam at 100 °C.

Which of the following occurs?

- 1 The forces between the molecules decrease.
- 2 The speed of the molecules increases.
- 3 The separation of the molecules increases.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

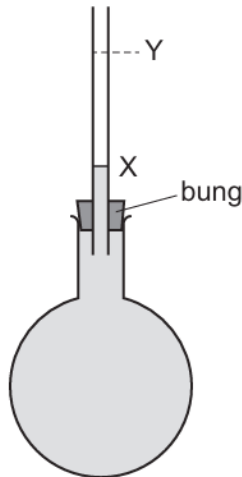
28. O/N/19/11/Q21

What is the name of the change of state that occurs when a substance changes from a solid to a liquid?

- A** boiling
B condensation
C evaporation
D melting

29. O/N/19/11/Q23

The diagram shows a glass flask full of water at 10 °C. A long glass tube passes through the bung into the water. The water level in the tube is at X.



The flask is placed in hot water and the water level in the tube eventually rises to Y.

Why does the water level rise?

- A** The flask contracts and the water expands.
B The flask contracts more than the water contracts.
C The flask expands and the water contracts.
D The flask expands less than the water expands.

30. O/N/19/11/Q24

A glass beaker contains water in a warm room.

The water is initially at the same temperature as the room.

Which change does **not** affect the initial rate of evaporation of the water?

- A** changing the temperature of the water
- B** using a beaker made of thicker glass
- C** using a beaker with a larger cross-sectional area
- D** using a fan to blow air over the beaker

31. O/N/19/12/Q18

A student makes four statements about evaporation and boiling.

- 1 Both involve an increase in molecular separation.
- 2 Both occur only at a fixed temperature.
- 3 Both involve the absorption of thermal energy.
- 4 Both occur throughout the liquid.

Which two statements are correct?

- A** 1 and 4 **B** 2 and 4 **C** 1 and 3 **D** 2 and 3

32. O/N/19/12/Q21

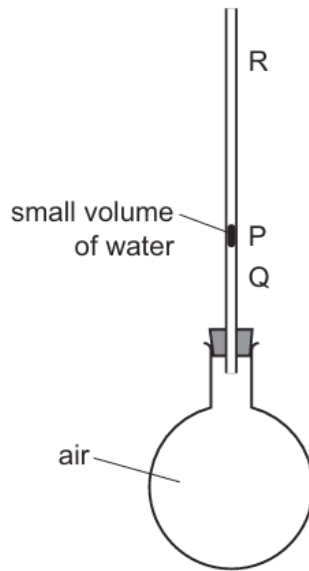
A metal disc is heated to 600 °C and then lowered into a beaker of water.

What happens to the mass of the metal disc and to the mass of the water in the beaker?

	mass of metal disc	mass of water in beaker
A	increases	decreases
B	increases	no change
C	no change	decreases
D	no change	increases

33. M/J/19/11/Q20

The diagram shows air trapped in a flask by a small volume of water in a thin tube.



When the flask is held in a student's hands, the small volume of water first moves down from P to Q, and then up to R.

Why does the small volume of water move like this?

- A The flask contracts and then the air expands.
- B The flask expands and then the air contracts.
- C The flask expands and then the air expands less than the flask.
- D The flask expands and then the air expands more than the flask.

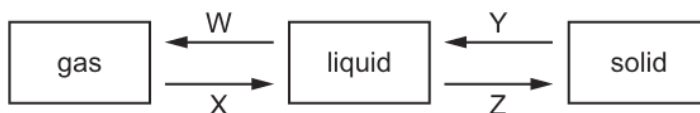
34. M/J/19/12/Q22

How is *latent heat* defined?

- A the thermal energy absorbed or emitted by unit mass during a change of state without a temperature change
- B the thermal energy absorbed or emitted by unit mass during unit temperature change
- C the thermal energy absorbed or emitted during a change of state without a temperature change
- D the thermal energy absorbed or emitted during unit temperature change

35. O/N/18/11/Q13

The diagram shows four changes of state W, X, Y and Z.



Which row gives the names of the four changes of state?

	W	X	Y	Z
A	condensation	boiling	freezing	melting
B	condensation	boiling	melting	freezing
C	boiling	condensation	freezing	melting
D	boiling	condensation	melting	freezing

36. O/N/18/11/Q14

What increases when a liquid becomes a gas at its boiling point?

- A** the average kinetic energy of the molecules
- B** the molecular size
- C** the molecular spacing
- D** the total number of molecules

37. O/N/18/11/Q15

A metal lid fits tightly on a glass jar.

Which process makes it easier to remove the lid from the jar?

- A** cool the lid only
- B** put the jar and lid in a refrigerator
- C** warm the jar only
- D** warm the lid only

38. O/N/18/11/Q16

The specific heat capacity of lead is $130 \text{ J/(kg } ^\circ\text{C)}$. Lead of mass 0.50 kg is heated from 10°C to 35°C .

Which calculation gives the amount of thermal energy, in J, absorbed by the lead?

- A** $0.50 \times 130 \times 25$
- B** $0.50 \times 130 \times 35$
- C** $\frac{0.50 \times 130}{25}$
- D** $\frac{0.50 \times 130}{35}$

39. M/J/18/11/Q17

What is described as 'the escape of more energetic molecules from the surface of a liquid'?

- A condensation
- B evaporation
- C melting
- D radiation

40. M/J/18/11/Q19

What happens when a solid is heated and expands?

- A The molecules do not change size and the spaces between the molecules become larger.
- B The molecules expand and the spaces between the molecules become larger.
- C The molecules expand and the spaces between the molecules become smaller.
- D The molecules expand and the spaces between the molecules do not change size.

41. O/N/17/11/O12

Scientists believe that some lakes are shrinking because of evaporation.

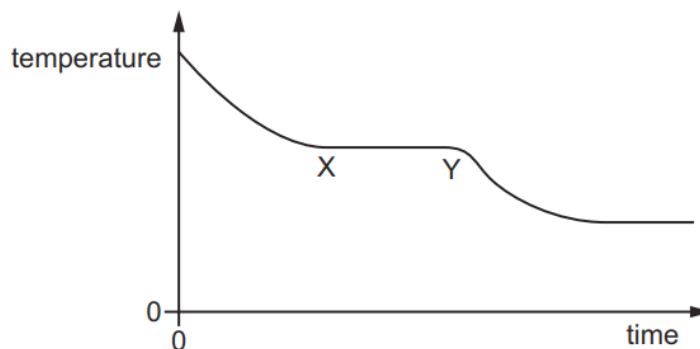
What increases the rate of evaporation?

- A a decrease in the surface area
- B a fall in water temperature
- C an increase in the depth of the lake
- D wind blowing across the surface

42. O/N/17/11/Q13

Solid wax is melted in a boiling tube and then allowed to cool.

The graph shows the cooling curve.



What is happening between points X and Y?

- A The liquid is at its boiling point.
- B The liquid is turning to solid.
- C The solid has reached room temperature.
- D The solid is turning to liquid.

43. O/N/17/11/Q14

During boiling and during melting, a substance either takes in energy or gives out energy.

Which row gives the energy transfer corresponding to the change of state?

	during boiling	during melting
A	gives out energy	gives out energy
B	gives out energy	takes in energy
C	takes in energy	gives out energy
D	takes in energy	takes in energy

44. O/N/17/11/Q15

The temperature of a 50 g mass of a metal is raised by 40 °C. The specific heat capacity of the metal is 0.40 J / (g °C).

How much thermal energy is supplied?

- A** 0.32 J **B** 0.50 J **C** 500 J **D** 800 J

45. O/N/17/12/Q13

A dish of liquid is left on a laboratory bench. Some of the liquid evaporates.

What happens and why?

- A** The liquid cools because liquid molecules have more potential energy than gas molecules.
B The liquid cools because faster-moving molecules escape.
C The liquid warms because liquid molecules have less potential energy than gas molecules.
D The liquid warms because slower-moving molecules are left behind.

46. O/N/17/12/Q15

What is meant by the *heat capacity* of a solid object?

- A** the maximum thermal energy it can store per unit mass
B the maximum thermal energy it can store per unit volume
C the thermal energy required to cause a unit temperature rise
D the thermal energy required to raise its temperature to its melting point

47. M/J/17/11/Q20

A substance consists of particles that are close together and moving past each other at random. The average speed of the particles is gradually increasing.

What best describes the substance?

- A** a gas being heated
- B** a liquid being heated
- C** a liquid undergoing solidification
- D** a solid being heated

48. M/J/17/11/Q21

The table shows the melting points and the boiling points of some elements.

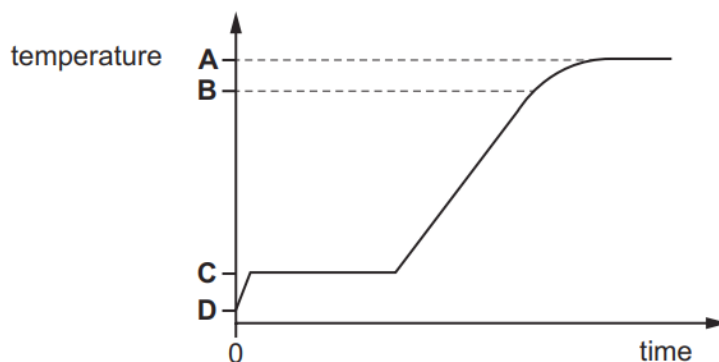
Which element is a liquid over the largest range of temperature?

	element	melting point /°C	boiling point /°C
A	bromine	-7	59
B	chlorine	-102	-34
C	fluorine	-220	-188
D	iodine	114	184

49. M/J/17/11/Q22

Some ice cubes are taken from a freezer and heated in a container. The readings of temperature and time are recorded on the graph.

Which temperature is 0 °C?



50. M/J/17/11/Q25

There is a tight metal lid on a glass jar. The lid is loosened by pouring hot water over it.

Which physics property explains this?

- A** convection
- B** evaporation
- C** expansion
- D** radiation

51. O/N/16/11/Q16

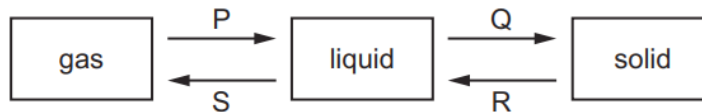
When a liquid evaporates, molecules escape from its surface.

Which molecules escape, and what happens to the average kinetic energy of the molecules remaining in the liquid?

- A** The less energetic molecules escape and the average kinetic energy decreases.
- B** The less energetic molecules escape and the average kinetic energy increases.
- C** The more energetic molecules escape and the average kinetic energy decreases.
- D** The more energetic molecules escape and the average kinetic energy increases.

52. O/N/16/11/Q17

The diagram shows changes of state.

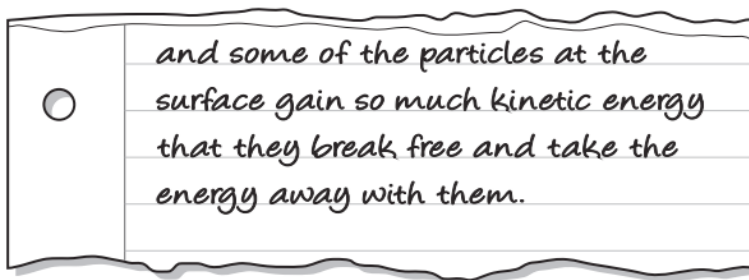


What are the names of the changes of state P, Q, R and S?

	P	Q	R	S
A	condensation	melting	solidification	boiling
B	condensation	solidification	melting	boiling
C	boiling	melting	solidification	condensation
D	boiling	solidification	melting	condensation

53. O/N/16/12/Q19

A piece of paper torn out of an exercise book is shown.



Which process is being described?

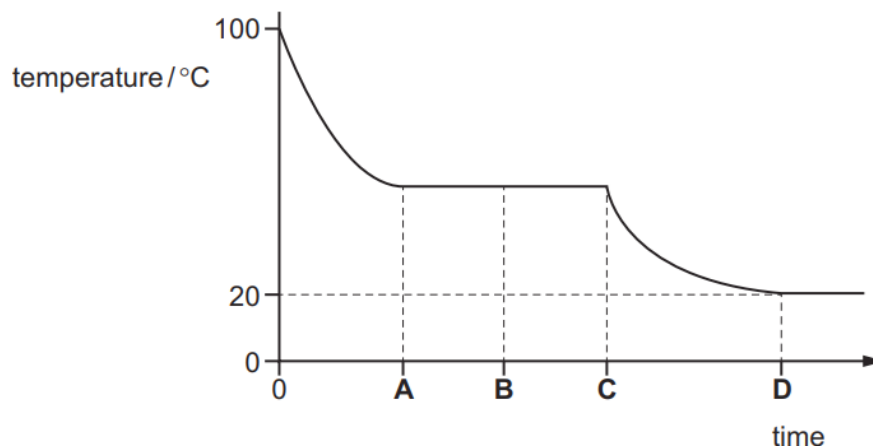
- A conduction
- B convection
- C evaporation
- D radiation

54. O/N/16/12/Q20

A sample of liquid at 100°C is allowed to cool to a room temperature of 20°C .

The diagram shows how the temperature changes with time during the cooling process.

At which time does the sample become completely solid?



55. O/N/16/12/Q21

One kilogram of water needs more thermal energy to raise its temperature by 1°C than one kilogram of copper.

Why is this?

- A copper has a higher specific heat capacity than water
- B copper has a lower specific heat capacity than water
- C copper is a better conductor of heat than water
- D copper is a poorer conductor of heat than water

56. O/N/15/11/Q17

An electric shower takes in cold water at 17°C . The shower gives 6000 J of energy every second to the cold water and heats it to 37°C . The specific heat capacity of water is $4200\text{ J}/(\text{kg }^{\circ}\text{C})$.

What is the mass of hot water supplied by the shower in one second?

- A** 0.035 kg **B** 0.039 kg **C** 0.071 kg **D** 0.084 kg

57. O/N/15/12/Q18

Some of the liquid in a dish evaporates, as shown in the diagrams.



Which molecules leave the liquid and which molecules in the liquid have greater average kinetic energy?

	molecules that leave have	molecules in the liquid have greater average kinetic energy
A	high energy	before evaporation
B	high energy	after evaporation
C	low energy	before evaporation
D	low energy	after evaporation

58. O/N/15/12/Q19

An insulated beaker contains 300 g of water, initially at 30°C . Water at 100°C is added until the temperature of the mixture reaches 50°C .

The specific heat capacity of water is $4.2\text{ J}/(\text{g }^{\circ}\text{C})$.

How much water is added?

- A** 60 g **B** 120 g **C** 180 g **D** 750 g

59. M/J/15/11/Q19

Which substance in the table is liquid at 20°C ?

	melting point $/^{\circ}\text{C}$	boiling point $/^{\circ}\text{C}$
A	-218	-183
B	-39	357
C	44	280
D	119	444

60. M/J/15/12/Q21

In a liquid, some energetic molecules break free from the surface even when the liquid is too cold for bubbles to form.

What is the name of this process?

- A** boiling
- B** condensation
- C** convection
- D** evaporation

61. M/J/15/12/Q22

A pupil adds 37 g of ice at 0 °C to 100 g of water at 30 °C. The final temperature of the water and melted ice is 0 °C. No heat is lost to, or gained from, the surroundings.

The specific heat capacity of water is 4.2 J/(g °C).

What is the specific latent heat of ice?

- A** 47 J/g **B** 341 J/g **C** 4700 J/g **D** 12 600 J/g

62. M/J/15/12/Q23

The heat capacity of an object, of mass 2.0 kg, is C . The energy needed to

- A** increase the temperature of the whole object by Δt is $C\Delta t$.
- B** increase the temperature of unit mass of the object by Δt is $C\Delta t$.
- C** melt the whole object is C .
- D** melt unit mass of the object is C .

63. O/N/14/11/Q18

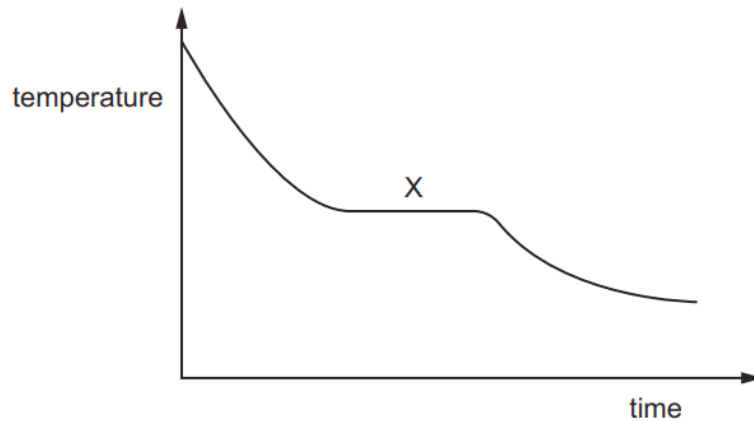
Water is poured into four dishes. In two of the dishes the water has a small surface area and in the other two it has a large surface area. The water in two of the dishes is cool and the water in the other two is warm.

From which dish does the water evaporate the quickest?

	surface area	temperature
A	large	cool
B	large	warm
C	small	cool
D	small	warm

64. M/J/14/11/Q20

A hot liquid is poured into a beaker. The graph shows how the temperature of the liquid changes as it cools towards room temperature.

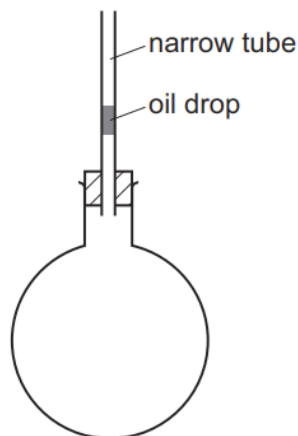


What is occurring at region X?

- A boiling and evaporation
- B condensation only
- C evaporation only
- D solidification and evaporation

65. M/J/14/12/Q21

The diagram shows a flask containing air. The air is trapped by a drop of oil in a narrow tube.



When the flask is heated the oil drop rises up the tube.

Which statement is **not** correct?

- A The air molecules each get larger.
- B The air molecules hit the container with greater force.
- C The air molecules move faster.
- D The air molecules move further apart.

66. M/J/14/12/Q22

Thermal energy is transferred to a solid. First it melts and then it boils to produce a gas.

Which statement about the temperature is correct?

- A** When melting and boiling the temperature does not change.
- B** When melting and boiling the temperature increases.
- C** When melting the temperature increases but when boiling the temperature stays the same.
- D** When melting the temperature stays the same but when boiling the temperature increases.

67. O/N/13/11/Q14

The temperature shown by a mercury-in-glass thermometer increases.

Which of the following is constant?

- A** the density of the mercury
- B** the internal energy of the mercury
- C** the mass of the mercury
- D** the volume of the mercury

68. O/N/13/11/Q17

When a person climbs out of a warm swimming pool on a hot summer day, he feels cold.

Why does this happen?

- A** The air has a high specific heat capacity.
- B** The air is a better conductor of heat than water.
- C** The water droplets increase his surface area.
- D** The water takes heat from his body to evaporate.

69. O/N/13/12/Q18

Different amounts of energy are supplied to copper blocks of different masses.

Which block experiences the greatest temperature change?

	mass of block / kg	energy supplied / J
A	0.1	200
B	0.2	200
C	0.4	600
D	0.8	400

70. O/N/13/12/Q21

The more energetic molecules in a liquid may escape from its top surface.

What is this process called?

- A** boiling
- B** Brownian motion
- C** convection
- D** evaporation

71. M/J/13/12/Q17

A liquid-in-glass thermometer contains mercury.

Which physical property of the mercury varies with temperature, enabling the thermometer to operate?

- A** mass
- B** melting point
- C** resistance
- D** volume

72. M/J/13/12/Q18

Thermal energy of 12 000 J is supplied to a 2.0 kg mass of copper.

The specific heat capacity of copper is 400 J / (kg °C).

What is the rise in temperature?

- A** 15 °C
- B** 30 °C
- C** 60 °C
- D** 100 °C

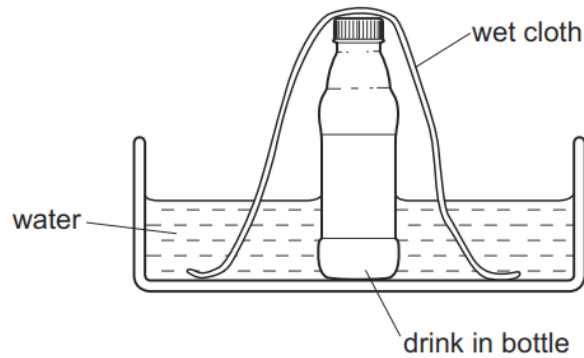
73. M/J/13/12/Q21

What happens to the molecules of a gas when the gas changes into a liquid?

- A** They move closer and lose energy.
- B** They move closer and gain energy.
- C** They move apart and lose energy.
- D** They move apart and gain energy.

74. O/N/12/11/Q16

On a hot day, the drink in a bottle can be kept cool by standing the bottle in a bowl of water and placing a wet cloth over it.

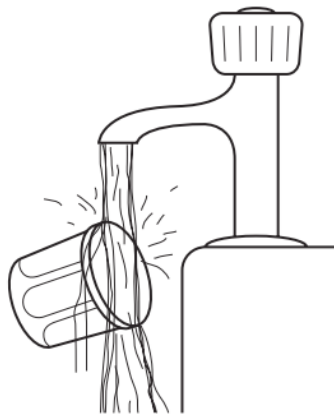


Why is the drink kept cool?

- A** Hot air cannot escape from the bottle.
- B** The cloth conducts heat from the bottle into the water.
- C** The drink cannot evaporate from the bottle.
- D** Water evaporating from the cloth cools the drink.

75. O/N/12/11/Q20

A person cannot unscrew the metal lid of a pot of jam. The lid can be unscrewed after it has been held under hot, running water for a few seconds.



Why is this?

- A** The air pressure in the jar falls.
- B** The glass expands.
- C** The jam melts.
- D** The metal lid expands.

76. O/N/12/12/Q16

In hot weather, people use electric fans to keep cool.

Why do the fans make them feel cool?

- A** They change one form of energy into another.
- B** They cool the air in the room.
- C** They increase the rate of evaporation from the skin.
- D** They speed up the vibration of air molecules.

77. O/N/12/12/Q20

A strip is made from two metals joined together. The diagrams show the strip at room temperature and after it has been cooled.



at room temperature



below room temperature

The change in shape occurs because

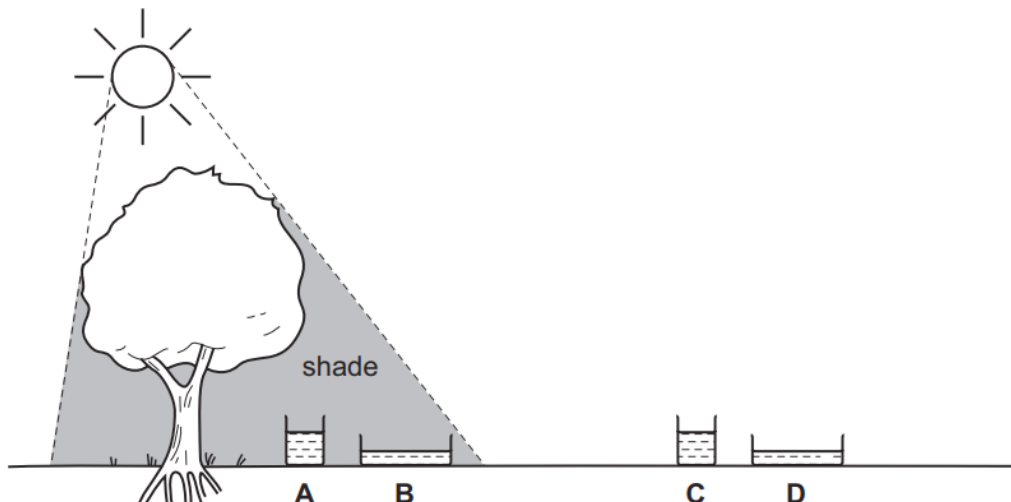
- A** brass contracts more than invar.
- B** brass expands when it cools down.
- C** invar and brass contract by equal amounts.
- D** invar contracts more than brass.

78. M/J/12/11/Q16

Four containers are filled with equal volumes of water at the same temperature.

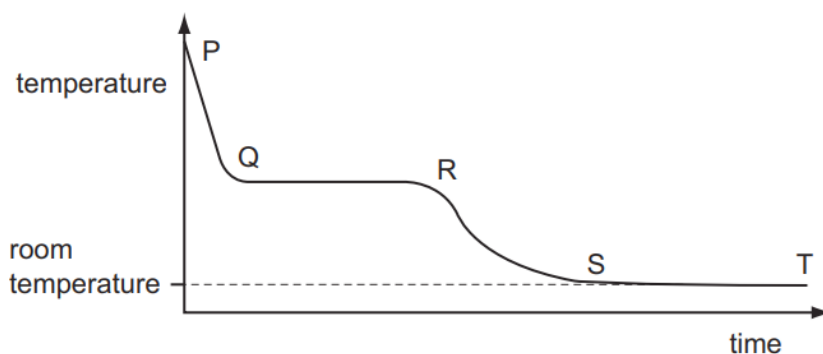
Containers **A** and **B** are shaded by a tree. Sunlight falls on containers **C** and **D**.

From which container does all the water evaporate first?



79. M/J/12/11/Q19

A hot liquid is allowed to cool. The graph shows the cooling curve.



In which part of the curve is latent heat released?

- A** PQ **B** QR **C** RS **D** ST

80. M/J/12/11/Q20

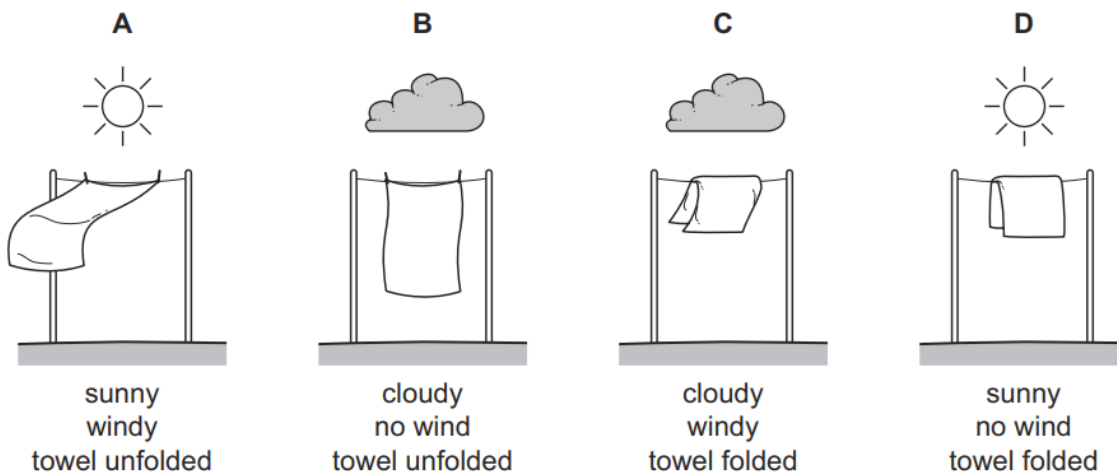
For the same temperature rise and the same original volume, which of the three states of matter expands the most and which state expands the least?

	expands most	expands least
A	gas	solid
B	liquid	gas
C	solid	gas
D	solid	liquid

81. M/J/12/12/Q16

Four wet towels are hung out to dry as shown.

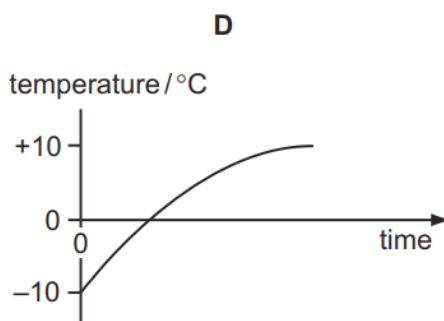
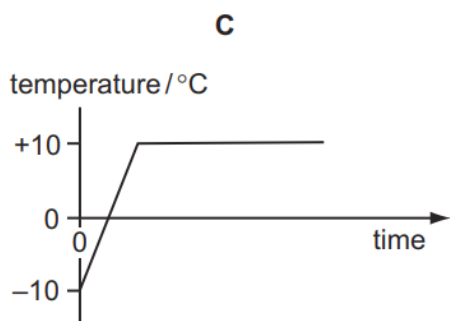
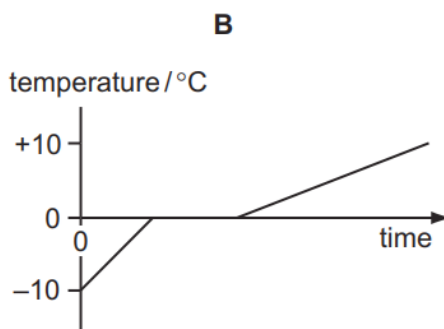
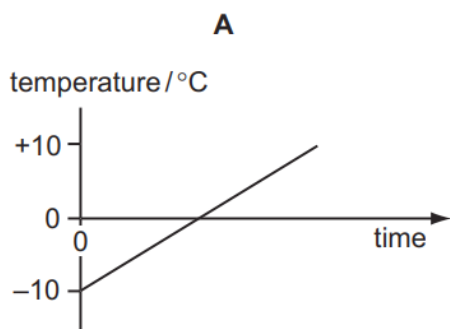
Which towel dries most quickly?



82. M/J/12/12/Q19

Ice at -10°C is heated until it is water at $+10^{\circ}\text{C}$.

Which graph shows how the temperature changes with time?



83. O/N/11/11/Q17

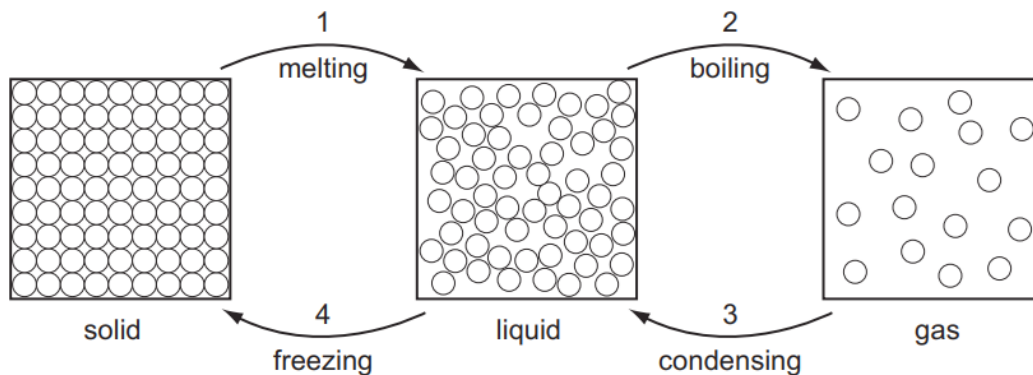
To raise the temperature of a 2.0 kg block of metal by 20°C , energy of 5.2 kJ is needed.

What is the value of the specific heat capacity of the metal?

- A** $0.13\text{ J}/(\text{kg } ^{\circ}\text{C})$
- B** $52\text{ J}/(\text{kg } ^{\circ}\text{C})$
- C** $130\text{ J}/(\text{kg } ^{\circ}\text{C})$
- D** $52\,000\text{ J}/(\text{kg } ^{\circ}\text{C})$

84. O/N/11/11/Q19

Substances can change from one state to another as shown.



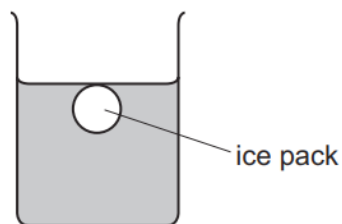
For substances to change from one state to another, there must be some energy transfer.

Which changes involve the substance taking in energy and which changes involve the substance giving out energy?

	energy taken in	energy given out
A	1 and 2	3 and 4
B	1 and 3	2 and 4
C	2 and 4	1 and 3
D	3 and 4	1 and 2

85. M/J/11/11/Q18

An ice pack is used to cool 0.25 kg of water. The specific heat capacity of water is $4.2 \text{ kJ}/(\text{kg}^\circ\text{C})$.



How much thermal energy (heat) must the ice pack extract from the water to reduce the water temperature by 15°C ?

- A** 0.070 kJ **B** 1.1 kJ **C** 16 kJ **D** 250 kJ

86. M/J/11/11/Q19

Fillings in teeth should be made from a material which

- A** does not expand when heated.
B expands by the same amount as the tooth when heated.
C expands less than the tooth when heated.
D expands more than the tooth when heated.

87. O/N/10/11/Q16

A liquid evaporates rapidly.

Why does this cause it to cool?

- A** Air molecules remove heat by contact with the liquid surface.
- B** Energy is lost by convection currents.
- C** Some of the most energetic molecules leave the liquid.
- D** The molecules have less room to move around.

88. O/N/10/11/Q20

What is caused by the thermal expansion of a substance when heated?

- A** a decrease in the resistance of a tungsten-filament lamp when switched on
- B** a rise in the pressure of the gas trapped in a gas cylinder placed in hot water
- C** the blowing of the fuse in a circuit when the current becomes too large
- D** the upward movement of the air above a Bunsen burner when it is lit

89. M/J/10/11/Q14

The liquid in a puddle evaporates and this causes its temperature to change.

How does the temperature of the liquid change and why?

	change	reason
A	decreases	less energetic molecules leave the liquid
B	decreases	more energetic molecules leave the liquid
C	increases	less energetic molecules leave the liquid
D	increases	more energetic molecules leave the liquid

90. O/N/09/Q16

A block of metal has a mass of 2.0 kg. Its specific heat capacity is $800 \text{ J/(kg } ^\circ\text{C)}$.

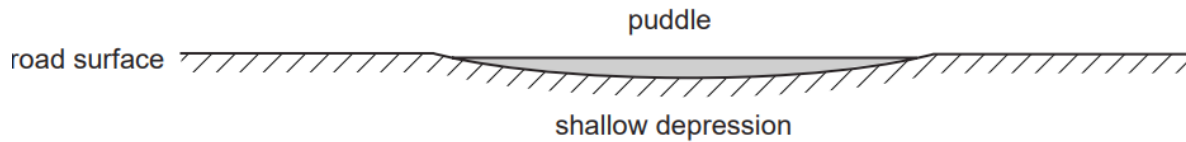
The block is supplied with 2400 J of energy.

What is the rise in temperature?

- A** 0.17°C **B** 0.67°C **C** 1.5°C **D** 6.0°C

91. O/N/09/Q18

The diagram shows a cross-section through a rain-water puddle formed in a shallow depression in a road surface.



Over a period of time, the air temperature, wind speed and wind direction all remain constant.

What happens to the rate of evaporation of water from the puddle?

- A** It decreases, because the surface area decreases.
- B** It increases, because the puddle gets shallower.
- C** It increases, because the surface area decreases.
- D** It remains constant.

92. M/J/09/Q17

What happens when a metal bar is heated?

- A** The distance between the molecules increases, making the bar longer.
- B** The molecules get larger, making the bar longer.
- C** The molecules vibrate more quickly, making the bar denser.
- D** The speed of the molecules increases, making the bar thinner.

93. O/N/08/Q20

When ice melts to become water, which force must be overcome?

- A** the attraction between electrons and the nucleus
- B** the attraction between the atoms in a molecule
- C** the force between molecules
- D** the force of gravity

94. O/N/08/Q21

Which factors increase the rate of evaporation of a liquid?

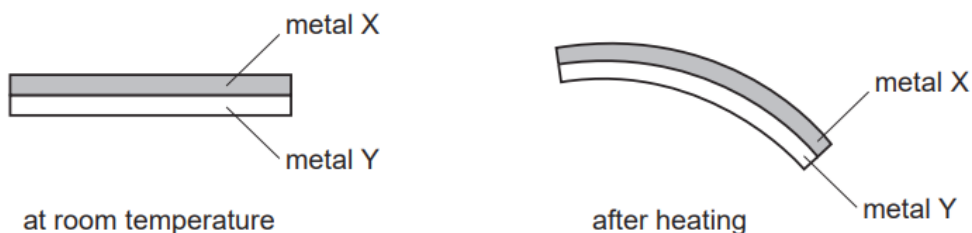
	increasing its temperature	increasing its surface area	increasing its depth
A	yes	yes	yes
B	yes	yes	no
C	yes	no	yes
D	no	yes	yes

95. O/N/08/Q19

The table shows the increase in length of four metals when heated through the same temperature rise. Each metal initially has the same length.

metal	increase in length/m
aluminium	0.000030
copper	0.000020
platinum	0.000009
steel	0.000010

A bimetallic strip is made from two of the metals. When heated, it bends in the direction shown.

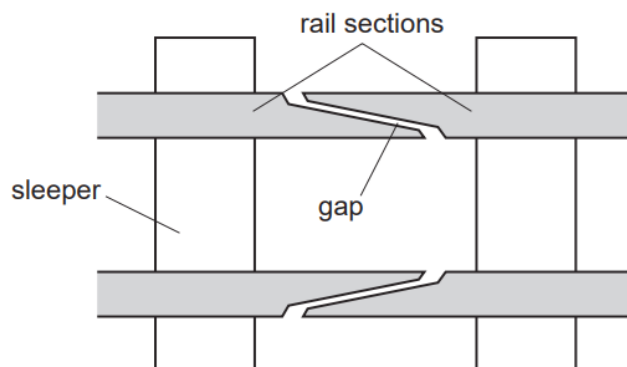


Which metals produce the above effect?

	metal X	metal Y
A	aluminium	platinum
B	copper	aluminium
C	steel	copper
D	platinum	steel

96. M/J/08/Q16

At regular intervals along a railway line there is a gap between the rail sections.



What is the reason for the gap between the rail sections?

- A** to allow for expansion of the rail sections during hot weather
- B** to allow for vibrations of the rail sections as the train passes over them
- C** to allow rain water to drain from the rail sections
- D** to keep the wheels of the train and carriages on the rail sections

97. M/J/08/Q17

Ice is taken from a freezer and left in a room. The ice melts and eventually the water reaches room temperature.

Which energy transfers take place?

	energy transfer during melting	energy transfer after melting
A	from ice to room	from water to room
B	from ice to room	from room to water
C	from room to ice	from room to water
D	from room to ice	from water to room

98. M/J/08/Q18

Some air is trapped inside a small balloon. The average kinetic energy of the air molecules in the balloon is increased.

What remains the same?

- A** the density of the air in the balloon
- B** the mass of the air in the balloon
- C** the temperature of the air in the balloon
- D** the volume of the air in the balloon

99. M/J/08/Q19

A student is investigating the evaporation of water.



The student can change:

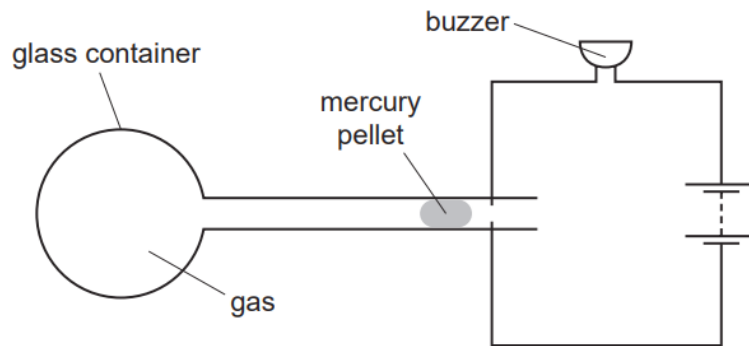
- the depth of the water;
- the surface area of the water;
- the temperature of the water.

How many of these changes, if any, would alter the rate at which evaporation occurs?

- A** 0
- B** 1
- C** 2
- D** 3

100. O/N/07/Q17

The diagram shows the design for an alarm.



The mercury pellet moves and completes the circuit.

Why does this happen?

	temperature	gas
A	falls	contracts
B	falls	expands
C	rises	contracts
D	rises	expands

101. M/J/07/Q13

A liquid is heated.

Which statement is **incorrect**?

- A The molecules expand.
- B The molecules gain energy.
- C The molecules move further apart.
- D The molecules move faster.

102. M/J/07/Q16

What is the definition of *heat capacity*?

- A the quantity of heat required to raise the temperature of an object through 1°C
- B the quantity of heat required to raise the temperature of 1 kg of a substance through 1°C
- C the quantity of heat required to convert an object from solid to liquid without a change in temperature
- D the quantity of heat required to change 1 kg of a substance from solid to liquid without a change in temperature

103. O/N/06/Q14

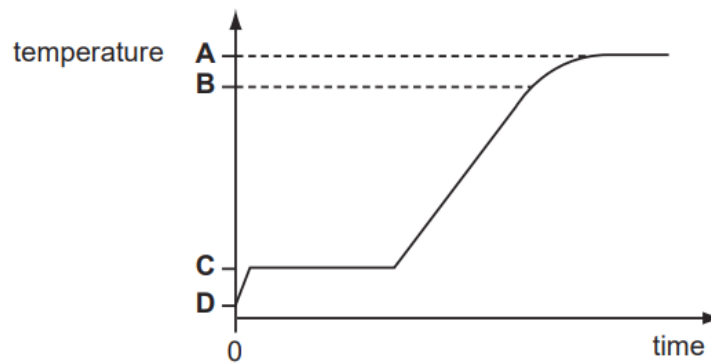
What conditions of temperature and surface area would produce the most rapid evaporation from a pool of water on a road surface?

	temperature	surface area
A	high	large
B	high	small
C	low	large
D	low	small

104. M/J/06/Q18

Some ice cubes are taken from a deep-freeze and placed in a metal container. The container is heated at a constant rate and readings of temperature and time are taken. The results are recorded on a graph.

Which temperature corresponds to 0°C ?

**105. M/J/06/Q19**

The diagrams show a bimetallic strip when it is at room temperature and after it has been cooled.



at room temperature



below room temperature

The change in shape occurs because

- A** brass contracts more than invar.
- B** brass expands when it cools down.
- C** invar and brass contract by equal amounts.
- D** invar contracts more than brass.

106. O/N/05/Q15

What will **not** affect the rate of evaporation from the surface of a liquid?

- A** depth of the liquid
- B** draughts above the surface of the liquid
- C** surface area of the liquid
- D** temperature of the liquid

107. M/J/05/Q14

Which factors increase the rate of evaporation of a liquid?

	increasing its temperature	increasing its surface area	increasing its depth
A	yes	yes	yes
B	yes	yes	no
C	yes	no	yes
D	no	yes	yes

108. M/J/05/Q16

In an experiment to find the specific heat capacity of a metal, it is found that 5200 J is needed to raise the temperature of a 2 kg block by 20 °C.

What value for the specific heat capacity is given by these results?

- A** 130 J/(kg °C)
- B** 520 J/(kg °C)
- C** 52 000 J/(kg °C)
- D** 104 000 J/(kg °C)

109. O/N/04/Q19

The fillings for a hole in a tooth should be made from a material that

- A** expands more than the hole in the tooth.
- B** expands by the same amount as the hole in the tooth.
- C** expands less than the hole in the tooth.
- D** does not expand when heated.

110. O/N/4/Q20

Heat energy is supplied at the same rate to 100 g of paraffin and to 100 g of water in similar containers.

Why does the temperature of the paraffin rise more quickly?

- A** The paraffin has a larger specific heat capacity than water.
- B** The paraffin has a smaller specific heat capacity than water.
- C** The paraffin is less dense than water.
- D** The paraffin is more dense than water.